Interpretation of Transformer FRA Responses—Part I: Influence of Winding Structure

Zhongdong Wang, *Member, IEEE*, Jie Li, *Student Member, IEEE*, and Dahlina M. Sofian, *Student Member, IEEE*

***Abstract*—Frequency-response analysis (FRA) has been accepted as one of the most sensitive tools to detect mechanical faults of power transformers. Correct interpretation of FRA responses is crucial when assessing the integrity of a transformer. Transformer FRA responses have distinctive frequency regions which are predominated by core, windings, and measuring setup. This paper addresses one of the major factors that affect the FRA responses: the winding structure itself. In terms of the structures of single winding, they can be categorized into windings with either high- or low-series capacitance in proportion to the shunt capacitance. Correspondingly, the FRA responses of windings of high series capacitance exhibit the increasing trend of magnitude, while windings of low series capacitance display a steady magnitude trend with the features of resonances and quasi-antiresonances or antiresonances. For a winding made of two concentric winding coils, the pattern of alternating resonances and antiresonances would be exhibited on its FRA response.**

***Index Terms*—Diagnosis, frequency-response analysis (FRA), interpretation, power transformers.**

I. Introduction

FREQUENCY-RESPONSE analysis (FRA) has been developed to detect winding displacement and deformation in power transformers [1]–[6]. FRA measurements are usually performed after faults or during periodical maintenance tests; by comparing the measured with the reference FRA response, any significant difference would indicate a potential mechanical problem of the winding, and the decision on the transformer would be made accordingly. However, apart from winding displacement and deformation, transformer design and manufacturing tolerance, the measuring setup or sometimes noises and disturbances can generate "normal" differences on FRA responses [1], [2], adding difficulties and complexities to FRA diagnosis.

The first step toward ensuring correct interpretation of the differences when comparing two FRA responses is to understand which factors govern explicit shapes and features of the FRA responses. Since FRA measurements are made at winding terminals across a wide range of frequencies, FRA responses tend to be divided into three different regions, from low to high frequencies, each of which is predominated by the transformer core, the windings, and the measuring setup. Clearly, the exact division of frequency regions may well depend on the transformer under test and the measurement equipment capacity. For example, conventional FRA measurement equipment, which uses their built-in 50-Ω measuring impedance and needs earthing leads with minimum length as the bushing height, will restrict the usefulness of the measured FRA response at high frequencies. It is therefore the authors' opinion that particular interests should be paid to the intermediate frequencies where the windings dominate. Indeed, the shape and features of an FRA response in the winding-dominant frequency region and, in particular, their changes, facilitate winding deformation diagnosis. In this aspect, the recent study carried out by Satish and Sahoo [7] is worth mentioning since they investigated the factors that affected the shape of the FRA responses in a double-wound transformer, although their simulation study was oversimplified and did not reflect the actual responses of transformer windings, it emphasized the importance of understanding the features of FRA responses for accurate FRA diagnosis.

In this paper, understanding FRA responses in relation to winding structures was developed by using experimental and simulation results of single continuous, intershielded, and interleaved disc-type windings. A winding can also be made of two concentric coils, either of the same type or a combination of two different structure types. An example of the former is the two-layer helical winding while the latter can be a main winding plus a tap winding. The effects of such a winding structure on FRA responses were studied through simulation by using the verified transformer model. For these windings, the factors that govern the shape and features of their FRA responses were then identified. It is recognized that although the structure of the winding under test dominates the features of its FRA response in the winding-dominant frequency region, mutual coupling between windings would also affect this frequency region and show themselves as certain features; consequently, the coupling issues between windings will be dealt with in latter parts of this series of papers.

II. Transformers and End-to-End FRA Measurements

In core-form power transformers, three common types of windings are employed, namely, the continuous, the intershielded, and the interleaved disc-type windings, while a single-layer (helical) winding structure is usually used for the tertiary. Continuous disc-type windings were popular in the early days due to the manufacturing simplicity. However, with the voltage ratings of the transformer increasing, the basic insulation level (BIL) was also increased. Coping with the nonlinearity of impulse voltage distribution along the winding

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The authors are with the School of Electrical and Electronic Engineering, The University of Manchester, Manchester, M60 1QD, U.K. (e-mail: zhongdong.wang@manchester.ac.uk).

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intershielded and interleaved disc windings superseded continuous disc windings in high-voltage power transformers.

With various winding structures employed, different FRA responses would be anticipated. Still, there should be some fundamental relationships that exist between the features of the FRA response and the winding structure since the frequency response of a winding is determined by the electrical characteristics of the winding equivalent circuit which is composed of the elements of resistance (R), inductance (L), and capacitance (C).

Although not standardized, FRA measurements are mainly carried out by injecting a voltage at one end and measuring the response voltage at the other end of the winding under test, while the untested winding terminals are left floating. This is the way in which the end-to-end FRA response of a winding is obtained [8], which is the same as the FRA winding "admittance" or "transadmittance" measurement called by different FRA users. All FRA responses in this paper were taken by using this measurement method unless otherwise stated. The measurement results were obtained by using an HP 4195A network analyzer, which is a two-channel 50-Ω equivalent impedance measurement device which generates a low-voltage (10 V) variable-frequency sinusoidal signal and measures the generated signal and the response signal of the test object, respectively. The frequency range for FRA measurements is conventionally set as being from 10 Hz to 2 MHz.

The measured response is usually shown graphically by plotting the logarithmic amplitude ratio of the response voltage to the input voltage in decibels (y scale) against the frequency (x scale). The frequency scale can be either logarithmic or linear and both have been used in practice; the logarithmic frequency scale is often used to show the complete frequency range, whilst the linear frequency scale is helpful when looking at discrete frequency bands and comparing small differences at particular frequencies.

III. Theoretical Analysis

A. Basic Features of End-to-End FRA Responses

At low frequencies, a single winding behaves as an inductive element, and the end-to-end FRA response follows a falling magnitude trend when the frequency is increased, with a linearly decreasing slope of approximately -20 dB per decade. The larger the core inductance is, the lower the FRA magnitude becomes. The magnetizing inductance L_m can be calculated as $L_m = N^2/R$, where N is the number of turns of the winding, and R is the reluctance of the magnetic path ($R = l/\mu A$, l is the length of the magnetic path, A is the cross sectional area of the core, and μ is the permeability). It is therefore not difficult to understand that FRA responses of a transformer with higher voltage and larger power ratings usually have a lower magnitude to begin with at low frequencies, and within the same transformer, the high-voltage (HV) winding response tends to have a lower magnitude than the low-voltage (LV) winding response due to the turn ratio. Also, it is understandable that for a three-phase transformer, the FRA response of the middle-phase winding at low frequencies is different from those of the other two phases, since the magnetic path of the middle phase is slightly different from those of the lateral phases due to the unsymmetrical magnetic core paths.

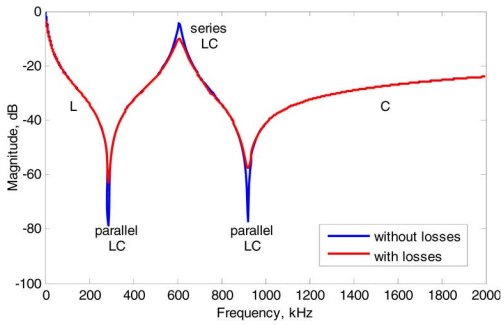


Fig. 1. Fundamental trends and features of FRA responses.

At high frequencies, a single winding behaves as a capacitive element due to the parallel path of winding series capacitance to the winding inductance being much smaller. However, it should be noted that a transformer has at least two windings per phase, and mutual inductive couplings between sections of the same winding or between windings exist and provide very complex current paths; therefore, resonances at high frequencies can also be observed in transformer FRA measurements. Consequently, there is a need to further study these high-frequency resonances for their use in FRA diagnosis and investigation of failure causes [9]. Taking a simplified theoretical approach and regarding a single winding largely as a capacitive element, the end-to-end FRA response follows a raising magnitude trend when the frequency is increased, with a linearly increasing slope of approximately 20 dB per decade. The larger the capacitance is, the higher magnitude that the FRA will have.

In the intermediate-frequency range, the combination of winding leakage inductance and series capacitance results in a paralleled LC. LC in parallel produces antiresonance at a certain frequency on the end-to-end FRA response, consequently blocking the signal at that particular frequency. This antiresonance signifies itself as a local minimum or so-called "valley" with a large negative decibel value.

The simplest representation of LC in series is a T connection, where the shunt capacitance is connected in the middle of the two halves of the winding inductance. The end-to-end FRA response of a T-connected LC network shows a series resonance, amplifying the signal at that particular frequency. This resonance signifies itself as a local maximum or so-called "peak" with a small negative decibel value.

The effect of frequency-dependent conductive and dielectric losses on the end-to-end FRA response is to attenuate the sharpness of the resonances and the antiresonances.

The basic trends and features of FRA responses as mentioned before are shown in Fig. 1. However, for a wide range of frequencies, the equivalent circuit of a winding includes multiple inductance and capacitance elements and moreover, there are mutual inductive and capacitive couplings between the winding elements, which effectively determine the features of FRA responses of the windings, including multiple resonances and antiresonances.

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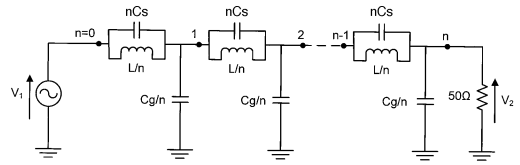


Fig. 2. n-stage lumped-element ladder network of the winding.

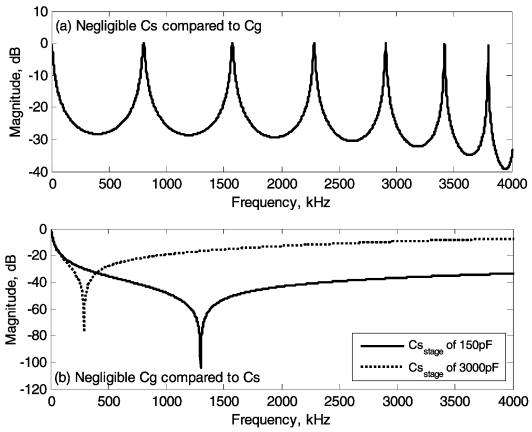


Fig. 3. FRA response of a single winding, measured on the equivalent network with two extreme cases: (a) $\mathrm{Cs} = 0$ and (b) $\mathrm{Cg} = 0$.

B. N-Stage Transformer Winding Lumped Ladder Network

A uniformly structured winding can be represented by an n-stage ladder network as shown in Fig. 2, with the winding leakage inductance L, the winding series capacitance C_s, and the shunt capacitance C_g evenly distributed between the n stages. The general solution for the voltage and current at any point x on the network can be represented by (1), where x is the number of stages along the winding, starting from the injecting end

$$\begin{aligned} u(x,j\omega) &= A\cosh(rx) + B\sinh(rx) \\ i(x,j\omega) &= \frac{1}{Z}[A\sinh(rx) + B\cosh(rx)] \\ \text{where, } r^2 &= \frac{-\frac{LC_g}{n^2}\omega^2}{1 - LC_s\omega^2} \text{ and } Z = \sqrt{\frac{L}{C_g(1 - LC_s\omega^2)}}. \end{aligned} \quad (1)$$

A and B are constants, Z is the characteristic impedance, and r is the propagation constant of the winding. At both ends of the winding, the boundary conditions for FRA measurements are

$$\begin{aligned} u(0,j\omega) &= A, \quad (x = 0) \\ u(n,j\omega) &= A\cosh(rn) + B\sinh(rn) = 0 \quad (x = n) \end{aligned} \quad (2)$$

where the approximation is made based on the assumption that the measuring impedance of 50 Ω at the response end of the winding is usually much smaller than the equivalent winding impedance. This assumption is reasonable for studying the features of resonances and antiresonances where the sensitivity of absolute magnitude measurement is not concerned.

The end-to-end FRA response can be derived

$$\frac{i(n,j\omega)}{u(0,j\omega)} = \frac{1}{Z}\frac{\sinh^2(rn) - \cosh^2(rn)}{\sinh(rn)} = \frac{-1}{Z\sinh(rn)}. \quad (3)$$

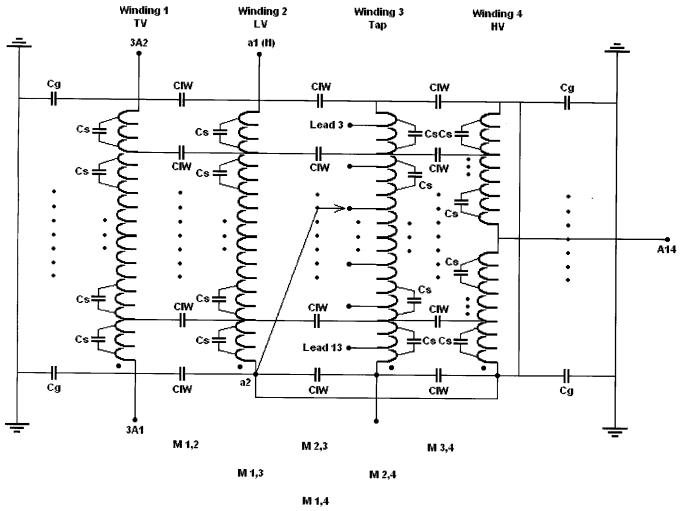


Fig. 4. Typical diagram of the transformer model used for simulation.

The end-to-end FRA response in (3) can then be divided into three frequency regions

$$\frac{i(n,j\omega)}{u(0,j\omega)} = \begin{cases} \frac{j}{Z\sin(|r|n)}, & f < \frac{1}{2\pi\sqrt{LC_s}} \\ \frac{1}{\infty} \approx 0, & f = \frac{1}{2\pi\sqrt{LC_s}} \\ \frac{1}{j|Z|\sinh(rn)}, & f > \frac{1}{2\pi\sqrt{LC_s}} \end{cases}. \quad (4)$$

In the frequency region where $f < 1/2\pi\sqrt{LC_s}$, although the end-to-end FRA response takes an inductive format in (4), $\sin(|r|\,n)$ will oscillate between [-1, 1] and force the response oscillating between capacitive and inductive trends. When $\sin(|r|\,n) = 0$, multiple local resonances are produced at the frequencies as

$$\begin{aligned} f_k &= \frac{k\pi}{2\pi\sqrt{LC_g + (k\pi)^2 LC_s}} \\ rn &= k\pi \quad k = 1,2,\ldots,n-1. \end{aligned} \quad (5)$$

Between two adjacent resonances, the magnitude of the response would reach a quasi-antiresonance where $|\sin(|r|\,n)| = 1$. At this local minimum, the magnitude is defined as $|1/Z| \neq 0$ and, therefore, the magnitude will not reach a large negative decibel value. Also, the shape of the response near this local minimum will be smooth and concave without the same sharp magnitude drop as the normal antiresonance. This series of quasi-antiresonances creates "artificial" local minimums or so called "troughs."

In (4), antiresonance will appear at $f = 1/2\pi\sqrt{LC_s}$, where the magnitude would drop sharply to the largest negative decibel value as the measuring system would allow. In the frequency region where

$$f > \frac{1}{2\pi\sqrt{LC_s}}$$

the magnitude will take an increasing trend and behaves capacitively.

From (5), it can be seen that as a winding behaves as an electric equivalent network made from multiple distributed parameters, the frequencies of the resonances are determined by the winding leakage inductance L, series capacitance C_s, and shunt

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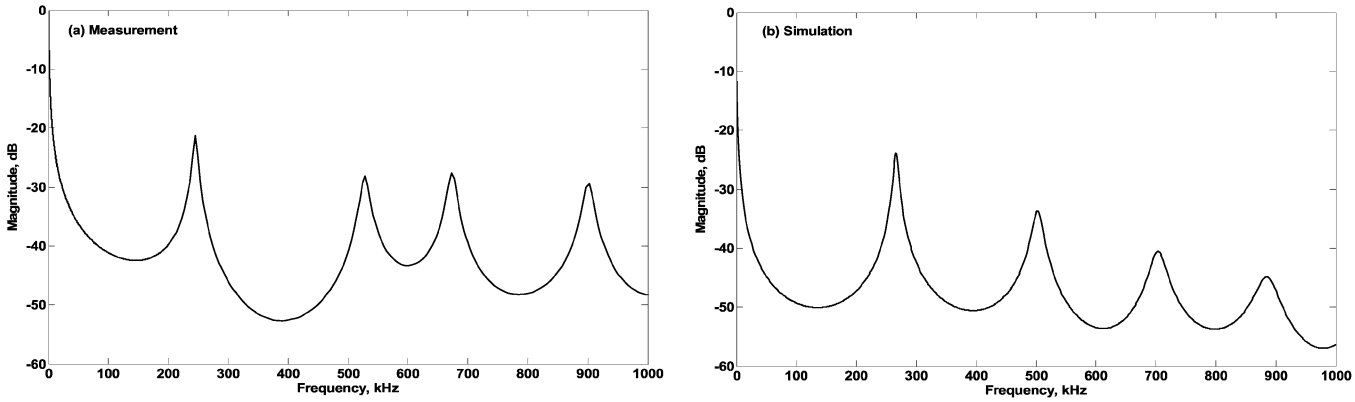


Fig. 5. Continuous disc winding (Cg $= 168$ pF, Cs $= 7$ pF, $\alpha = 4.89$).

capacitance C_g. There are two extreme cases with (5), when winding series capacitance is small enough to be neglected (i.e., $C_s = 0$), there will be resonances with even frequency intervals as $f_k = k/2\sqrt{LC_g}, \quad k = 1, 2, \ldots, n-1$. On the other hand, when winding series capacitance is extremely large and makes C_g seem almost nonexistent, say $C_g = 0$, there will be no resonance at all. Fig. 3 shows the measured FRA responses of a laboratory eight-stage lumped ladder network made of LC components to represent two types of windings: windings with low- or high- series capacitance. This network has an equivalent winding inductance $L = 800\ \mu$H, an equivalent winding series capacitance $C_{\text{sLOW}} = 19$ pF, or $C_{\text{sHIGH}} = 375$ pF, and an equivalent shunt capacitance $C_g = 480$ pF. The extreme cases of this lumped network with negligible C_s or C_g depict the features of winding with low series capacitance: a series of "peaks" and "troughs" with steady magnitudes in Fig. 3(a) and the features of winding with high series capacitance: little resonances existing and an explicitly capacitive increasing trend in Fig. 3(b). Some FRA practitioners also call the features in Fig. 3(a) a "U"-shape feature which would be used in this paper hereinafter.

This theoretical study suggests that, in general, the proportion of the series capacitance C_s and the shunt capacitance C_g is significant in determining the FRA response for a specific winding structure. It does not only determine the magnitude trend but also the shapes and the positions of resonances and antiresonances.

IV. FRA Simulation Model

A complete lumped-element model of the transformer can be constructed by using a series of n-stage ladder networks as shown in Fig. 4. In this equivalent circuit, the unit of the lumped element is formed of a double disc for disc-type windings or a turn for layer-type windings. The inductance and the capacitance of the unit are calculated by using winding geometry and material properties, the frequency-dependent conductive and dielectric losses are also included as resistance. To ensure the accuracy of the model, mutual capacitive and inductive couplings between units are carefully considered in the calculation to be included in the model.

The inductance matrix includes the core and the leakage inductance, whereas the capacitance matrix includes the shunt, the series, and the interwinding capacitance. The system solution of the equivalent network is obtained when applying Kirchoff's voltage and current law (KVL and KCL) at each node of the network. Through solving the network equations, the simulated FRA response of the windings is obtained. All FRA responses onwards in the simulation studies will be produced by using this modeling methodology.

V. Single Windings

Single-layer windings and the continuous disc windings have lower series capacitance, while the intershielded disc windings and the interleaved disc windings tend to have higher series capacitance.

Two homogeneous windings (one continuous and the other interleaved disc winding) and an inhomogeneous winding—a partially intershielded + continuous disc winding—were studied.

The comparison between the measured FRA responses on these windings and the corresponding simulated responses are shown in Figs. 5–7, respectively. The measured and simulated FRA responses are matched reasonably well; moreover, the features of the continuous, the intershielded, and the interleaved disc windings are shown in the simulation results.

It should be kept in mind that the purpose of transformer modeling and FRA simulation is never to provide "exact" matches with measured results; instead, it is developed to aid the understanding and interpretation of the measured results, especially when used to identify the features which are indicative for winding displacement and deformation.

The values of the winding shunt (Cg) and series capacitances (Cs) are calculated, also the spacing coefficient $\alpha\left(\alpha = \sqrt{C_g/C_s}\right)$ during the simulation, in order to analyze the FRA response of the winding structure by using the knowledge developed earlier through the theoretical studies.

A. Continuous Disc Winding

The continuous disc winding used in this study is a 6.6-kV winding of a 1-MVA mine transformer. The FRA response of this winding is characterized by a series of well-defined resonances and quasi- antiresonances with a steady magnitude trend as shown in Fig. 5. The winding Cs $= 7$ pF is relatively low as

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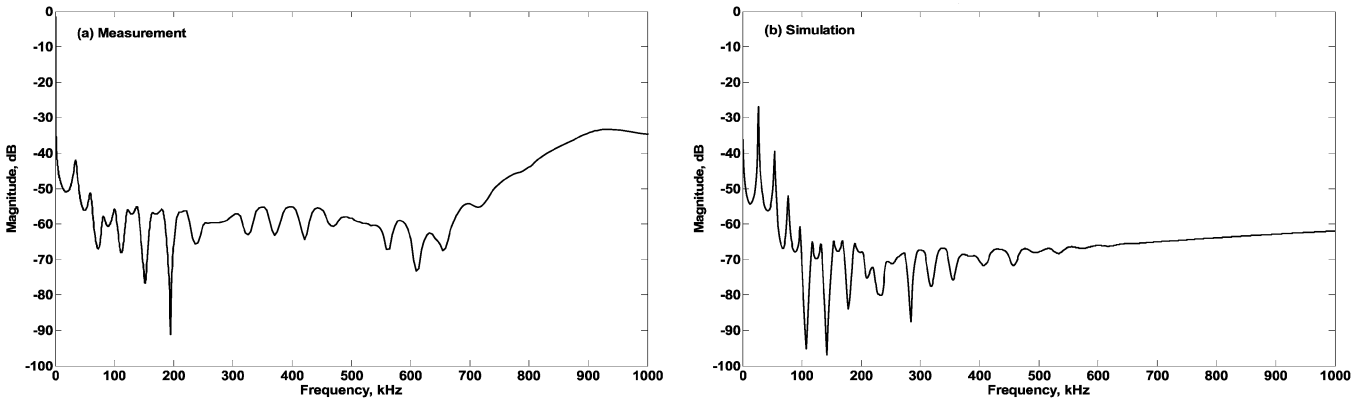


Fig. 6. Intershielded disc winding (Cg $= 1010$ pF, Cs $= 40$ pF, $\alpha = 5.02$).

opposed to Cg $= 168$ pF, resulting in space coefficient $\alpha = 4.89$.

Since this continuous disc type of winding is from a distribution transformer with a low-power rating and has a low series capacitance, its uniform winding structure gives the FRA response the exact 'U' shape feature exhibited as in Fig. 3(a).

B. Intershielded Disc Winding

The intershielded disc winding used in this study is of a generator transformer winding design; it is an inhomogeneous winding, with almost a half winding structured as intershielded discs and the other as continuous discs. In Fig. 6, the measured and simulated FRA results show the same features at the frequencies up to 600 kHz. Above 600 kHz, the measured FRA response exhibits a faster increasing trend in magnitude than the simulation result. This can be due to the fact that the series capacitance is calculated to be lower than the actual value or that the inductive effect of the earthing lead is introduced in the measuring setup.

It is observed in Fig. 6 that the FRA response of the intershielded disc winding initially starts with a series of "U"-shaped resonances and quasi-antiresonances, followed by the "camel hump"-shaped response with damped resonances and antiresonances. These antiresonances are caused by the high series capacitance of the intershielded part of the winding. High series capacitance in the intershielded part of the winding also causes the damped resonances. The response looks like an inverted "U" shape following the "camel humps" in the high frequencies where the losses are high. Below 600 kHz, the steady trend of magnitude exists for the FRA response. Features shown in Fig. 6 have been also observed on FRA measurements of LV windings of various generator transformers and tertiary windings of transmission transformers.

This type of winding structure has similar properties to the continuous disc-type winding, as the winding series capacitance (Cs $= 40$ pF) is also relatively low compared to the winding shunt capacitance (Cg $= 1010$ pF). The space coefficient $\alpha = 5.02$ is also similar to the previous exemplar continuous disc winding.

The space coefficient is a global parameter which reflects the overall ratio between the total winding shunt and series capacitance and does not reflect the local improvement of nonlinearity through partial use of intershielded discs. Yet, the differences in the FRA responses of these two types of windings are distinctive when considering the features of sharp antiresonances of the "camel hump" with the intershielded disc winding. In this respect, the FRA response would be a better indicator for identifying different types of winding structures which have nearly the same space coefficient α.

C. Interleaved Disc Winding

The interleaved disc winding is a purpose-made laboratory winding which consists of 28 discs with 22 turns per disc. In addition, static rings are connected at the top and the bottom of the winding.

Due to the high series capacitance of interleaved disc windings, their FRA responses would exhibit less resonances of which most are at low frequencies [7], [10], [11]. This is also demonstrated clearly in Fig. 7, which looks similar to Fig. 3(b) with negligible Cg. The FRA response begins with antiresonance, followed by the increasing trend of magnitude caused by the dominant high winding series capacitance (Cs $= 278$ pF) as compared with the shunt capacitance (Cg $= 144$ pF). This winding has a space coefficient $\alpha = 0.72$.

The comparable shunt and series capacitance values of interleaved disc winding make its space coefficient α smaller ($\alpha \sim 1$) than those of the other continuous and intershielded disc windings ($\alpha \sim 5$). The aforementioned studies give an insight on the shape and features of FRA responses in relation to the single winding structure under test.

If a winding is presented with mechanical problems, the changes in its physical dimension would result in changes of electrical parameters; given sufficient changes, they would then alter the features of the FRA responses.

VI. Windings Having Two Concentric Coils

In practice, a winding under test may well constitute more than one coil; for example, a winding with taps for voltage regulation is usually made of two concentric coils in a high-voltage power transformer. Therefore, the effect of the interaction between the main winding with the tap winding needs to be considered. Also, sometimes a generator transformer LV winding can be made of two concentric helical coils (a two-layer helical winding), and the interaction between the two coils would affect

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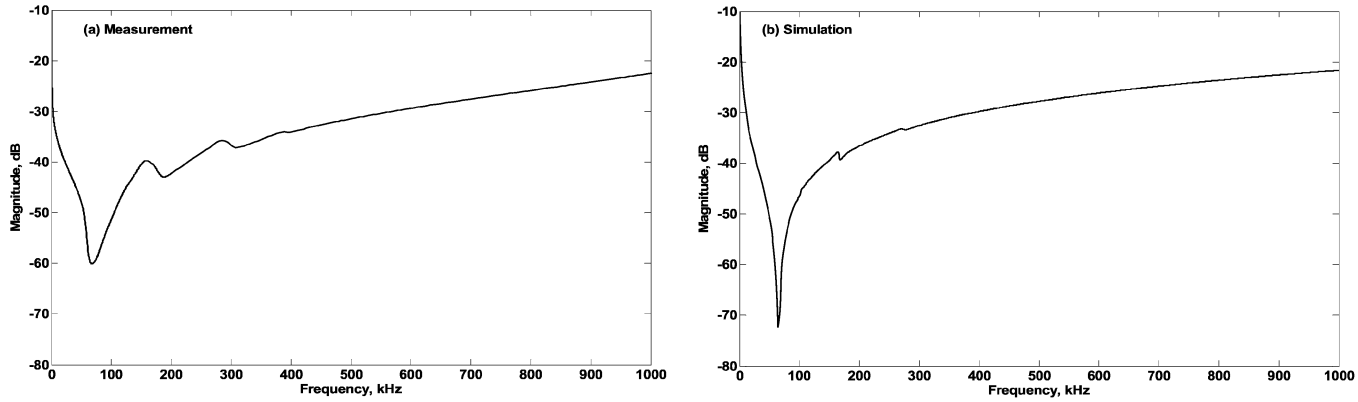


Fig. 7. Interleaved disc winding ($\mathrm{Cg} = 144$ pF, $\mathrm{Cs} = 278$ pF, $\alpha = 0.72$).

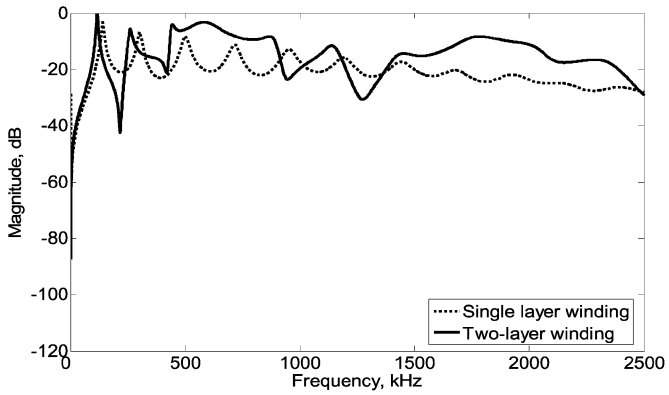


Fig. 8. Comparison between FRA responses of a two-layer winding ($\mathrm{Cg} = 6353$ pF, $\mathrm{Cs} = 102$ pF, $\alpha = 7.9$) and a single-layer winding ($\mathrm{Cg} = 6397$ pF, $\mathrm{Cs} = 5.9$ pF, $\alpha = 32.8$).

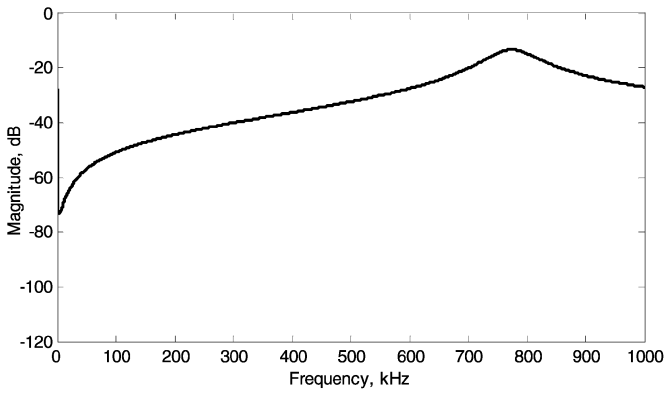


Fig. 9. FRA response of the multistart tap winding ($\mathrm{Cg} = 898$ pF, $\mathrm{Cs} = 2877$ pF, $\alpha = 0.6$).

the winding FRA response. This section of this paper will use the simulation results to discuss the features of FRA responses of these types of winding structures.

A. Two-Layer Helical Winding

A 13-kV tertiary winding can be designed as either a single layer or a two-layer helical winding, both winding structures are modeled and the simulation results are shown in Fig. 8. The FRA response of the single-layer winding shows the typical "U" shape for a winding with low series capacitance, which corresponds well with the calculated space coefficient $\alpha = 32.8$. On the other hand, the FRA response of the two-layer helical winding shows a distinctively different pattern with alternating resonances and antiresonances.

Comparing the winding series capacitance, the single-layer winding has a very low $\mathrm{Cs} = 5.9$ pF while the two-layer winding has a $\mathrm{Cs} = 102$ pF. This is because that between the two concentric layer coils, the intercoil capacitance effectively contributes to the winding series capacitance, which causes the occurrence of the sharp antiresonances shown in Fig. 8.

B. Intershielded Disc Main Winding With Tap Winding

An LV winding in a 240-MVA 275/132-kV transmission transformer includes a main intershielded disc winding and a multistart tap winding, which are electrically connected and physically concentrically arranged. The FRA measurement was taken under the maximum tap position.

A multistart-type winding is similar to a vertically interleaved disc winding, only much simpler in structure but still having high series capacitance. The calculated shunt capacitance of this tap winding is $\mathrm{Cg} = 898$ pF, and the series capacitance $\mathrm{Cs} = 2877$ pF, resulting in a space coefficient $\alpha = 0.6$. The FRA response of the multistart tap winding shown in Fig. 9 is dominated by the increasing trend in magnitude across the frequency range until a distinct resonance occurs at a frequency between 700–800 kHz.

When the multistart tap winding is connected in series with the main LV intershielded disc-type winding, the FRA response of the LV winding at the maximum tap position is compared with that of the main LV winding as shown in Fig. 10.

In Fig. 10, the main LV winding case has the features of the FRA response of a single intershielded disc winding as shown in Fig. 6, while the combination case of the LV with the tap winding has the pattern of alternating resonances and antiresonances, which can be also seen in Fig. 8. Similarly, the intercoil capacitance between the LV and the tap windings contributes effectively to the equivalent winding series capacitance, resulting in $\mathrm{Cs} = 1600$ pF, which also smoothes the response above 500 kHz into a capacitive increasing trend of magnitude. In this two-coil combination case, the calculated space coefficient is $\alpha = 3.4$.

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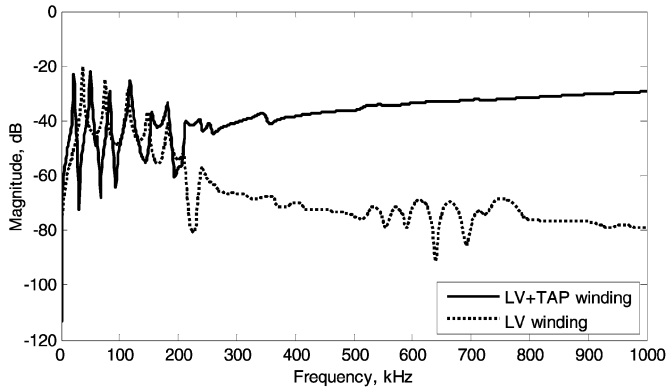


Fig. 10. Comparison of the FRA responses between the main intershielded disc LV winding (Cg = 1708 pF, Cs = 13 pF, $\alpha = 11.4$) and the combination of this LV winding plus the multistart tap winding (Cg = 1803 pF, Cs = 160 pF, $\alpha = 3.4$).

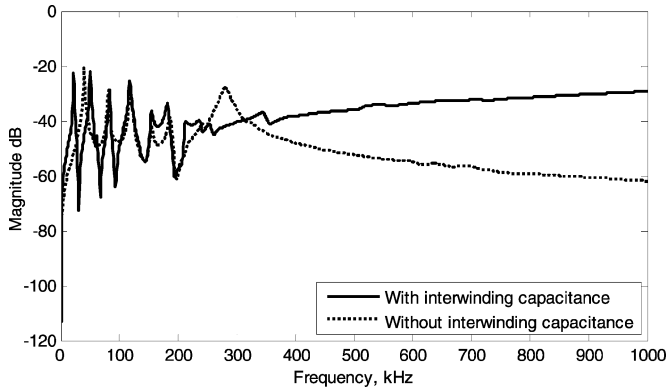


Fig. 11. Comparison of FRA responses obtained with (Cg = 1803 pF, Cs = 160 pF, $\alpha = 3.4$) and without (Cg = 1803 pF, Cs = 13 pF, $\alpha = 11.8$) intercoil capacitance cases for an LV intershielded disc-type winding connected with a multistart tap winding.

In order to further confirm the point that it is the intercoil capacitance between the LV disc winding and the multistart tap winding that contributes mostly toward the alternating resonances and antiresonances, the FRA response of the combination of the LV and the tap winding was remodeled and simulated only with the change that the intercoil capacitance is set to zero as nonexistence. In a practical sense, this could be achieved by shielding. The new simulation result is shown in Fig. 11.

As shown in Fig. 11, by taking off the intercoil capacitance, the features of alternating resonances and antiresonances have disappeared and the capacitive increasing trend of magnitude has also been replaced by the inductive decreasing trend of magnitude in the frequency range above 400 kHz. Without intercoil capacitance, the resulting space coefficient for the two-coil combination case is $\alpha = 11.8$.

VII. Conclusion

The features of the FRA responses for differently structured windings were investigated in this paper through simulation.

In terms of the structures of a single winding, they can be categorized into windings with either high- or low- series capacitance in proportion to the shunt capacitance. Correspondingly, the FRA responses of windings of high series capacitance exhibit the increasing trend of magnitude across the frequency range with little resonances and antiresonances. A good example of windings with high series capacitance is an interleaved disc type of winding. The windings of low series capacitance display the steady magnitude trend with the features of resonances and quasi-antiresonances or antiresonances. As an example, for a single layer or a continuous disc winding, quasi- antiresonances are mostly likely to occur whereas the intershielded disc-type windings tend to have quasi-antiresonances followed by antiresonances, making so called "U" shape or "camel hump" shape features.

For a winding made of two concentric coils, the alternating resonances and antiresonances pattern exhibits on the FRA responses. Simulation studies show that the main contribution toward this pattern is due to the intercoil capacitance effectively increasing the equivalent winding series capacitance.

Apart from the structure of the winding under test, the interaction between the winding under test and nontested windings in the same phase, and the interaction between different phases due to delta connection, would affect the measured FRA responses. Their impact on the FRA responses would be the topic of discussion in the latter parts of this series of papers through modeling the complete three-phase transformer configuration.

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Zhongdong D. Wang (M'00) was born in China in 1969. She received the B.Eng. and M.Eng. degrees in high-voltage engineering from Tsinghua University, Beijing, China, in 1991 and 1993, respectively, and the Ph.D. degree in electrical engineering from the University of Manchester Institute of Science and Technology, Manchester, U.K., in 1999.

She is a Senior Lecturer with the Electrical Energy and Power Systems Group in the School of Electrical and Electronic Engineering of the University of Manchester. Her research interests include condition monitoring, transformer modeling, oil/paper aging mechanism, and alternative insulation materials.



Jie Li (S'07) was born in China in 1979. He received the B.Eng. degree in electrical engineering and automation from Xi'an Jiaotong University, Xi'an, China, in 2002 and the M.Sc. degree in electrical power engineering from the University of Manchester Institute of Science and Technology, Manchester, U.K., in 2005, and is currently pursuing the Ph.D. degree in electrical energy and power systems from the School of Electrical and Electronic Engineering at the University of Manchester, Manchester, U.K.

His research interests include power transformer high-frequency modeling, fast transients, and transformer fault diagnosis.



Dahlina M. Sofian (S'07) was born in Malaysia in 1979. She received the B.Eng. degree in electrical power engineering from the University of Southampton, Southampton, U.K., in 2001 and the M.Sc. degree in electrical power engineering from the University of Manchester Institute of Science and Technology, Manchester, U.K., in 2002, where she is currently pursuing the Ph.D. degree in electrical power engineering with the Electrical Energy and Power Systems Group in the School of Electrical and Electronic Engineering at the University of Manchester, Manchester, U.K.

Her research interests include condition monitoring and diagnosis techniques for power transformers.

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